

Trajectory-Resolved Modular Charge Response: A Finite-Size CPU Pilot

Asadullah Bhuiyan¹

¹*Department of Physics, Cornell University, Ithaca, New York 14853, USA*

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We analyze saved trajectory covariance histories of hard and soft programmable domain walls using two state-resolved probes: a retarded charge response under modular-time evolution and a static modular susceptibility. The 20×20 histories contain ten independent trajectories, and every nonlinear modular construction is performed before averaging translated cuts or trajectories. The hard wall exhibits endpoint-selective, oppositely directed response on its two physical interfaces. A finite-window Fourier transform resolves mirror-related branches with $\omega \simeq b + v_F|k|$ and $v_F \simeq 0.28$ in opposite momentum sectors. One soft-wall branch has the same slope, while its partner is unresolved. The small longitudinal window, nonzero and damping-sensitive intercept, and strong aperture replicas make this an estimator pilot, not a controlled conformal-dispersion measurement.

I. DATA AND ESTIMATOR ORDER

The two legacy histories use $N_x = N_y = 20$, $n_{\text{shell}} = 1$, perfect correction, $\alpha_1 = 1$, $\alpha_2 = 30$, raster- y serial order, and a maximally mixed initializer. The hard protocol has `dw_truncation=true` and `meas_slab_only=true`; the soft protocol has both controls false at execution. Each history contains $S = 10$ independent trajectories and every covariance from cycle 0 through 40. We analyze cycles 21, 25, 29, 33, 37, 40 and all 20 translated half-cylinders

$$A(y_0) = [0, N_x] \times [y_0, y_0 + N_y/2], \quad y_0 = 0, \dots, 19. \quad (1)$$

For every observable we first evaluate the nonlinear estimator for each cut, average the 20 cuts within a trajectory, and only then form trajectory means and percentile-bootstrap intervals. Translated cuts and checkpoints are never treated as independent samples.

II. GAUSSIAN MODULAR RESPONSE

The repository stores the centered covariance G_A , so the restricted correlation matrix is

$$C_A = \frac{\mathbf{1}_A + G_A}{2} = U \text{diag}(\nu_a) U^\dagger. \quad (2)$$

For the Gaussian reduced density operator

$$\hat{\rho}_A = Z_A^{-1} \exp \left[- \sum_{ij} (h_A)_{ij} \hat{c}_i^\dagger \hat{c}_j \right], \quad (3)$$

the one-body modular Hamiltonian is

$$h_A = \log \frac{\mathbf{1}_A - C_A}{C_A} = U \text{diag}(\varepsilon_a) U^\dagger, \quad (4)$$

$$\varepsilon_a = \log \frac{1 - \nu_a}{\nu_a}.$$

We clip occupations to $[\epsilon, 1 - \epsilon]$, with primary $\epsilon = 10^{-10}$ and controls $10^{-8}, 10^{-12}$.

A. Retarded response

Let P_0 project onto both local orbitals of the perturbed unit cell. The many-body charge-phase kick $e^{-i\lambda\hat{N}_0}$ induces the correlation-matrix tangent

$$X(0) = -i[P_0, C_A]. \quad (5)$$

Since $[h_A, C_A] = 0$, modular-time evolution gives

$$X(\tau) = e^{-ih_A\tau} X(0) e^{ih_A\tau}, \quad \chi^R(\mathbf{r}, \tau) = \text{tr}[P_{\mathbf{r}} X(\tau)]. \quad (6)$$

In the modular eigenbasis,

$$\tilde{X}_{ab}(\tau) = -i(\nu_b - \nu_a)(\tilde{P}_0)_{ab} e^{-i(\varepsilon_a - \varepsilon_b)\tau}. \quad (7)$$

The spatial sum of Eq. (6) vanishes. We therefore use the signed wall dipole rather than a normalized center of mass,

$$M_y(\tau) = \sum_{\substack{x: d(x, x_w) \leq 2 \\ y_{\text{rel}}}} (y_{\text{rel}} - y_{\text{src}}) \chi^R(x, y_{\text{rel}}, \tau). \quad (8)$$

The early-time slope v is fitted over $0.1 \leq \tau \leq 0.8$.

B. Static susceptibility

For $h_\lambda = h_A - \lambda P_0$ and $f(x) = (1 + e^x)^{-1}$, the exact Fréchet derivative is

$$(\delta C_A)_{ab} = -f^{[1]}(\varepsilon_a, \varepsilon_b)(\tilde{P}_0)_{ab}, \quad (9)$$

where

$$f^{[1]}(x, y) = \begin{cases} [f(x) - f(y)]/(x - y), & x \neq y, \\ -f(x)[1 - f(x)], & x = y. \end{cases} \quad (10)$$

The local static susceptibility is $\chi^{\text{stat}}(\mathbf{r}) = \text{tr}[P_{\mathbf{r}} \delta C_A]$. Positive λ lowers the local modular chemical potential and attracts charge.

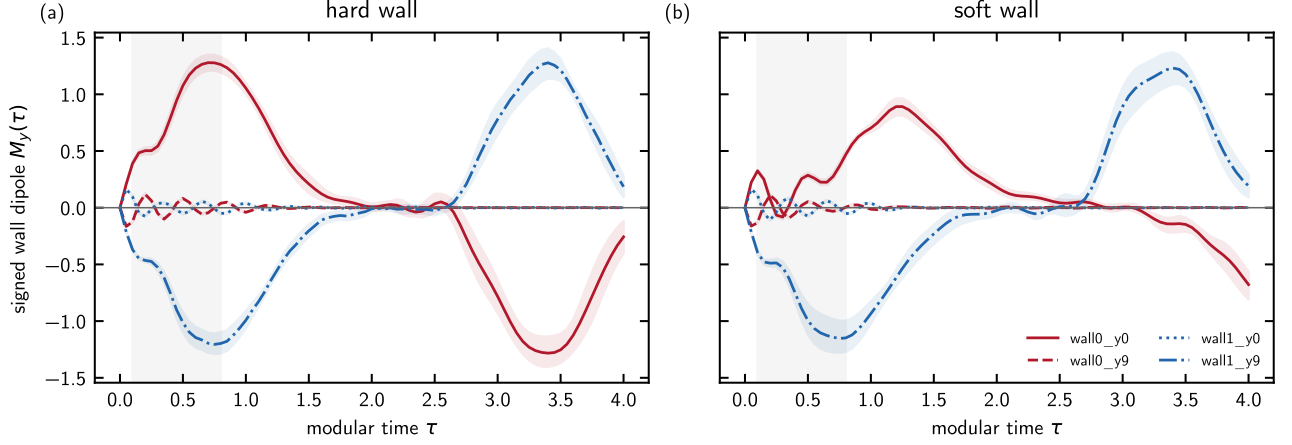


FIG. 1. Trajectory-resolved signed wall dipoles at cycle 40 for the hard and soft wall protocols. Curves are means over $S = 10$ independent maximally-mixed-initialized trajectories after averaging the 20 translated cuts within each trajectory; bands are trajectory-bootstrap 95% intervals. The shaded interval is the slope-fit window $0.1 \leq \tau \leq 0.8$. One endpoint on each physical wall carries the dominant response, with opposite signs between wall orientations.

III. REAL-SPACE RESPONSE

At cycle 40 the active source slopes are

$$(v_0, v_1)_{\text{hard}} = (1.514, -1.417), \quad (11)$$

$$(v_0, v_1)_{\text{soft}} = (0.384, -1.286), \quad (12)$$

where 0 denotes $(x, y_{\text{rel}}) = (5, 0)$ and 1 denotes $(15, 9)$. The complementary endpoints are at least an order of magnitude weaker. As an exploratory orientation-aligned summary, define

$$H = \frac{v_0 - v_1}{2}. \quad (13)$$

Trajectory bootstraps with 5000 resamples give

$$H_{\text{hard}} = 1.466 [1.350, 1.575], \quad (14)$$

$$H_{\text{soft}} = 0.835 [0.715, 0.943], \quad (15)$$

$$H_{\text{hard}} - H_{\text{soft}} = 0.630 [0.474, 0.795]. \quad (16)$$

The endpoint pair entering H was selected after inspecting all four source responses; accordingly, H is a descriptive handedness score, not a prespecified hypothesis test or quantized transport coefficient.

The static response is strongly local. Its source-averaged integrated value is 0.1352 for the hard wall and 0.1259 for the soft wall, while transverse leakage is of order 10^{-3} . It acts here as a local modular-compressibility diagnostic, not a chirality estimator. The maximum retarded total-charge residual is 1.04×10^{-15} . Analytic retarded and static responses agree with centered finite differences at relative errors 1.67×10^{-9} and 1.43×10^{-9} , respectively.

IV. FINITE-WINDOW FOURIER RESPONSE

Projecting Eq. (6) onto the five-column wall window gives $\chi_w(y, \tau)$. We compute

$$\chi_w(k, \omega) = \Delta\tau \sum_{j,y} e^{i\omega\tau_j} e^{-ik(y-y_{\text{src}})} w(\tau_j) e^{-\eta\tau_j} \chi_w(y, \tau_j), \quad (17)$$

with a Tukey taper, primary damping $\eta = 0.35$, ten independent momenta $k_n = 2\pi n/10$, and $0 \leq \tau \leq 4$. Because the half-cylinder is open along y_{rel} , Eq. (17) is a source-resolved finite-window transform rather than a momentum-conserving susceptibility.

For each wall orientation, we select its low-frequency momentum sector: $k < 0$ for $(5, 0)$ and $k > 0$ for $(15, 9)$. At the four nonzero points with $0 < |k| \leq 2.6$, the largest $|\chi_w|$ peak in $0.2 \leq \omega \leq 2.5$ is fitted to

$$\omega_{\text{peak}} = b + v_F |k|. \quad (18)$$

A branch is rejected when at least half of its peaks remain on the lower frequency-search boundary.

The resolved branches share $v_F \simeq 0.28$ and the hard-wall pair appears in opposite physical momentum sectors. This is consistent with a linear chiral modular branch. It

TABLE I. Crude Fourier-ridge fits at cycle 40. Intervals are trajectory-bootstrap 95% intervals. The η range reports fits at $\eta = 0.20, 0.35, 0.50$.

branch	v_F	ensemble R^2	η range
hard (5, 0)	0.279 [0.270, 0.290]	0.999	0.266–0.318
hard (15, 9)	0.282 [0.274, 0.291]	0.999	0.267–0.309
soft (5, 0)	unresolved	—	—
soft (15, 9)	0.284 [0.277, 0.294]	0.999	0.271–0.321

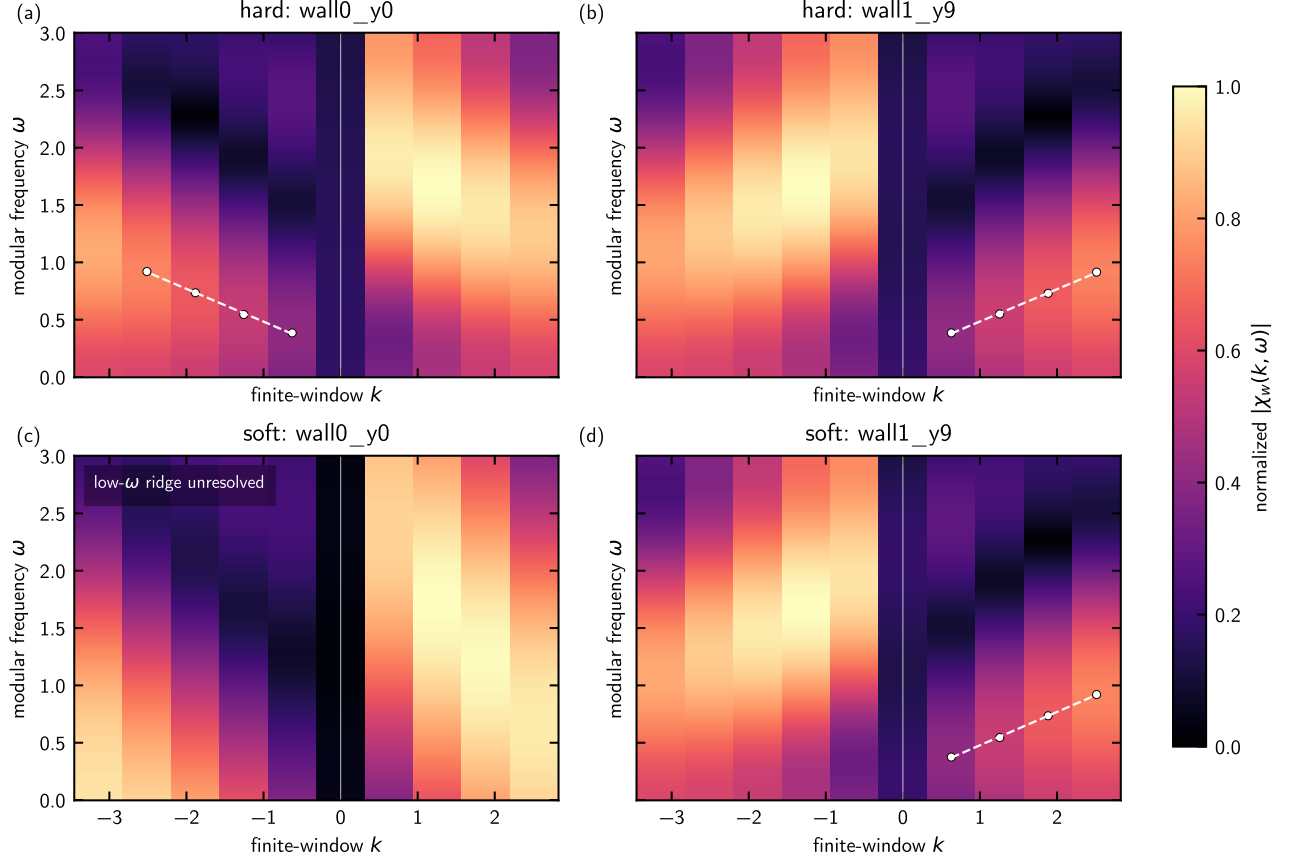


FIG. 2. Normalized finite-window Fourier magnitude at cycle 40. The four independent low-frequency ridge points and linear fits are overlaid where the branch passes the boundary check. The hard-wall branches occupy opposite momentum sectors and are mirror related. The soft (15,9) branch is resolved, whereas the soft (5,0) response piles up at the low-frequency search boundary. Every panel is formed per trajectory before averaging $S = 10$ trajectories.

is not yet a controlled $\omega = vk$ result: only four momenta enter each fit, the primary intercept is $b \simeq 0.19\text{--}0.20$, b moves strongly with damping, and finite-aperture replicas are prominent in Fig. 2. In particular, the soft wall does not yield a clean two-interface dispersion at this size.

V. SCOPE AND REPRODUCIBILITY

This pilot validates the response estimators and motivates a larger- N_y study, but it does not establish a conformal velocity, central charge, level, or manuscript gate. The ten trajectories use a maximally mixed initializer and are not a substitute for the redesigned random-pure W1 ensemble. The executed notebook, raw response arrays, Fourier product, summary tables, and figures are stored together under `notebooks/modular_response_cpu_pilot/`. The source covariance archives remain immutable and are addressed by SHA-256 digest in the notebook manifest.